

1. Introduction to Green's Theorem

We now introduce a useful tool for solving problems in electrostatics. This tool is useful for solving boundary value electrostatic problems. We are talking about the Green's Theorem and all its variations.

We know Gauss's divergence theorem which can be stated as follows

$$\iiint_V \vec{\nabla} \cdot \vec{A} d^3\vec{r} = \oiint_S \vec{A} \cdot d^2\vec{r}$$

$$\text{or, } \iiint_V \vec{\nabla} \cdot \vec{A} d^3\vec{r} = \oiint_S \vec{A} \cdot \hat{n} da$$

Let us take $\vec{A} = \varphi \vec{\nabla} \psi$ where φ and ψ are two scalar functions.

We can now calculate

$$\vec{\nabla} \cdot (\varphi \vec{\nabla} \psi) = \varphi \vec{\nabla} \cdot (\vec{\nabla} \psi) + (\vec{\nabla} \varphi) \cdot (\vec{\nabla} \psi)$$

and

$$\varphi (\vec{\nabla} \psi) \cdot \hat{n} = \varphi \frac{\partial \psi}{\partial n}$$

Using the above relations on the Gauss's divergence theorem we have

$$\iiint_V (\varphi \vec{\nabla} \cdot (\vec{\nabla} \psi) + (\vec{\nabla} \varphi) \cdot (\vec{\nabla} \psi)) d^3\vec{r} = \oiint_S \varphi \frac{\partial \psi}{\partial n} da$$

As φ and ψ are arbitrary scalar functions we can interchange them and similarly obtain the following equation

$$\iiint_V (\psi \vec{\nabla} \cdot (\vec{\nabla} \varphi) + (\vec{\nabla} \psi) \cdot (\vec{\nabla} \varphi)) d^3\vec{r} = \oiint_S \psi \frac{\partial \varphi}{\partial n} da$$

Subtracting the previous two equations we obtain the following

$$\iiint_V (\varphi \vec{\nabla} \cdot (\vec{\nabla} \psi) - \psi \vec{\nabla} \cdot (\vec{\nabla} \varphi)) d^3\vec{r} = \oiint_S \left\{ \varphi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \varphi}{\partial n} \right\} da$$

The equation

$$\iiint_V (\varphi \nabla \cdot (\nabla \psi) + (\nabla \varphi) \cdot (\nabla \psi)) d^3 \vec{r} = \iint_S \varphi \frac{\partial \psi}{\partial n} da$$

is known as *Green's first identity* and the equation

$$\iiint_V (\varphi \nabla \cdot (\nabla \psi) - \psi \nabla \cdot (\nabla \varphi)) d^3 \vec{r} = \iint_S \left\{ \varphi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \varphi}{\partial n} \right\} da$$

is known as *Green's second identity*.

In the following classes we will solve some example problems and it will become clearer how these identities are used in electrostatics.